INCH-POUND

MS27637B
29 September 2000
SUPERSEDING
MS27637A
18 December 1987

DETAIL SPECIFICATION SHEET

ELBOW, FLARELESS, TUBE TO HOSE-45°, SWIVEL NUT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83296.

REQUIREMENTS:

1. Elbows shall be in accordance with figure 1 and table I.

2. Material shall be a corrosion resistant steel, class 304 per SAE AMS5639, 304L per SAE AMS5647, 316 per SAE AMS5648, or 321 per SAE AMS5645, or condition A (accepted classes only), as specified in SAE AMS-QQ-S-763.

3. Finish shall be passivated as specified in MIL-DTL-83296.

4. Radius or break all corners to .005 + .005/-.000, unless otherwise specified.

5. All burrs and slivers shall be removed.

6. Surface roughness, in accordance with ASME B46.1, shall not exceed 125 µin R_a, unless otherwise specified.

7. All tolerances shall be ±.005, unless otherwise specified.

8. The part number and manufacturer's part number shall be marked in accordance with MIL-DTL-83296.

AMSC N/A FSC 4730

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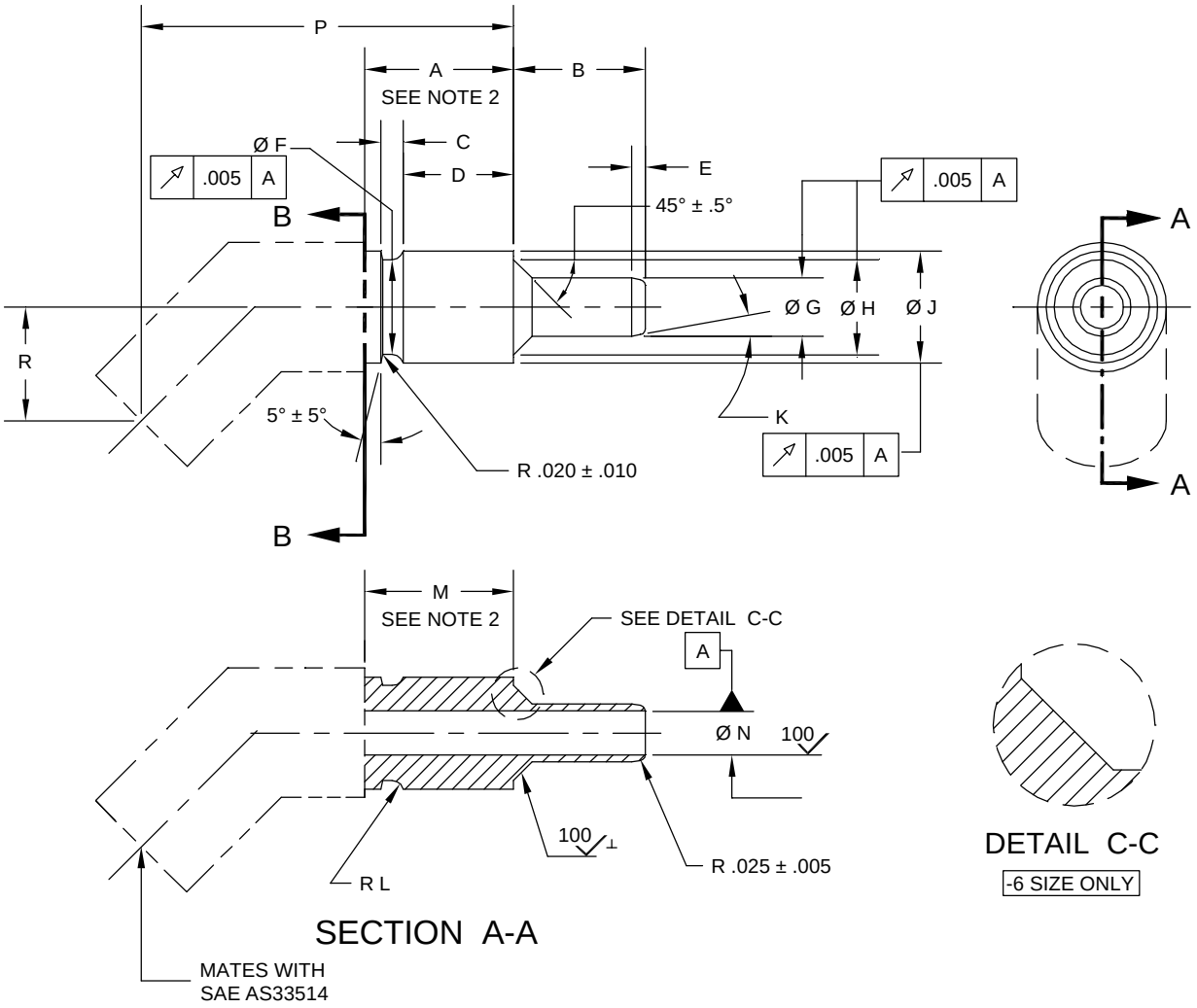

FIGURE 1. Flareless, 45°, Elbow.

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TABLE I. Elbow dimensions.

Part number	A Min	B ±.010	C		D +.005 -.000	E ± .005	F +.000 -.005
MS27637-4C	.633	.550	.116	±.010	.413	—	.346
MS27637-6C	.709	.620	.140	±.020	.442	—	.395
MS27637-8C	.779	.737			.472	—	.564
MS27637-10C	.660	1.050	.170		.405	.150	.679

TABLE I. Elbow dimensions - Continued.

Part number	G +.000 -.005	H ±.005	J ±.003	K ±.50°	L ±.004
MS27637-4C	.250	.420	.443	—	.040
MS27637-6C	.343	—	.497	—	.047
MS27637-8C	.445	.610	.680	—	
MS27637-10C	.588	.762	.823	10°	.062

TABLE I. Elbow dimensions - Continued.

Part number	M Min	N ±.005	P		R	
MS27637-4C	.578	.161	1.519	±.020	.515	±.020
MS27637-6C	.654	.271	1.840		.663	
MS27637-8C	.692	.375	2.179		.761	
MS27637-10C	.606	.465	1.824	±.035	.756	±.062

NOTES:

1. All dimensions are in inches, unless otherwise specified.

2. On -4, -6, and -8 size use dimension A when the adjacent diameter to the left of plane B-B is greater than dimension J. When the adjacent diameter is equal to or less than dimension J, dimension M may be used in place of dimension A.

3. On -10 size the bent tube, before forming, shall not be less than the nipple I.D.. Ovality of all bent tube fittings shall not exceed 7.5% of nominal tubing O.D..

4. Any forged design fitting is acceptable for -4, -6, and -8 size and bent tube design fitting is acceptable for -10 size.

5. This part is for use with MIL-DTL-83298 hose.

6. This specification sheet takes precedence over documents referenced herein.

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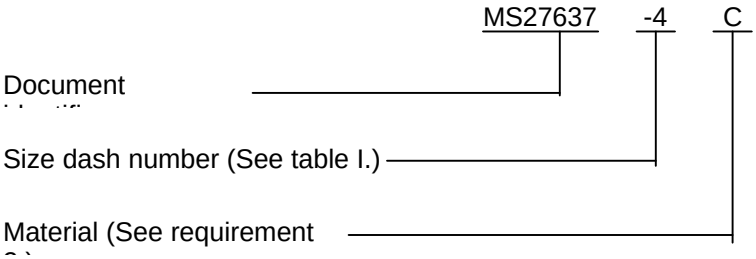
7. This is a design specification sheet for manufacturing purposes, the item is only procured as on integral part of an adapter assembly.

8. Referenced documents shall be of the issue listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

9. Dimensioning and tolerancing shall be in accordance with ASME Y14.5M-1994.

10. Part or Identifying Number (PIN):

Example of a PIN: MS27637-4C.



11. Next assembly is MS27633.

Custodians:
Air Force - 99
Army - AV
Navy - AS
DLA - CC

Review activities:
Air Force - 82
Navy - MC, SA

Preparing activity:
DLA - CC

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